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REGIONAL TOPICS FROM EUROPE

Life cycle assessment of coke production based on real plant and model-optimised data

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Abstract

Purpose Coke production is essential for the iron and steel industry, but this process can release several pollutants into the air, soil, and water, which cannot be ignored. Moreover, coke oven gas purification requires additional energy and auxiliary materials, resulting in significant indirect emissions. The purpose of this study to assess the environmental impact of coke production based on the data of a real coking plant and compare four different operation scenarios using model optimised data.

Method The "cradle to gate" life cycle assessment of the investigated coking plant were conducted by subdivision method. The functional unit was determined as the production of 1 ton of coke. Four scenarios were compared in this research work. The first one assumed normal operating conditions (scenario 1), the second one assumed environmentally optimised conditions as defined by the steady-state model of the gas purification technology (scenario 2), the third one assumed energetically optimised conditions as also defined by the model (scenario 3), and the fourth one assumed no gas purification technology at all (scenario 4). The study was performed using Sphera's Life cycle assessment software (formerly known as GaBi) based on ISO 14040 and ISO 14044 standards, and nine environmental impact categories were considered.

Results and discussion Besides coke production, transportation and gas purification had the most highest relative impact in most categories, with an average of 32%, 20%, and 12%, respectively. Among the input auxiliary materials, electricity contributed the most to the investigated categories (avg. 62%), followed by steam (avg. 26%) and process water (avg. 11%). The comparative analysis revealed that, compared to the base scenario, the energetically optimised scenario had the lowest environmental impact in the most considered categories (e.g. GWP decreased from 546.54 to 544.15 kg CO_2 eq). The optimisation reduces the energy demand of the plant, while the pollutant emissions do not exceed the standards; therefore, the direct emissions do not lead to a significant increase in environmental impact. The sensitivity analysis shows that a 10% reduction in electricity consumption significantly decreases the environmental impact of all the sub-processes.

Conclusions This is the first study to perform a coke production life cycle assessment using optimised data calculated by a process simulator alongside real plant data. Through this, the intention was to provide valuable insights into the environmental significance of coke oven gas cleaning technology.

Keywords Life cycle assessment · Coke production · Coke oven gas · Gas purification · Real plant data · Optimised data

1 Introduction

Coke is one of the most essential raw materials for pig iron production (Babich and Senk 2019). Coke is produced by heating the coking coal mixture at high temperatures in air-tight chambers, while coke oven gas (COG) is formed as a by-product. The gas's quantity and composition depend on the volatile content of the coal blend and the temperature used in the coke oven chambers (Loison et al. 1989; Manka 1979). Raw coke oven gas contains high-value components and contaminants (Bermúdez et al. 2013). After separation, valuable products such as coal tar or light oil (mainly

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benzene, toluene and xylene fraction) can be merchandised (Towsey et al. 2013). The concentration of polluting components in the gas should be removed as much as possible or at least reduced to below environmental standards since the main impurities (H_2S and NH_3) cause corrosion of equipment and pipelines (Caillat 2017) and produce SO_x and NO_x during combustion (Yen et al. 2015). The purified coke oven gas contains the highest percentage of hydrogen and methane (Zhang et al. 2020), so it has almost as high heating value as natural gas (Razzaq et al. 2013). Coke oven gas is, therefore, an excellent substitute for natural gas and can provide a significant part of the energy needs of integrated iron and steel plants (Sui et al. 2023). In addition, various on-site and off-site application methods have been examined, including combined heat and power (CHP) plants, methanol synthesis, hydrogen, ammonia and syngas production (Razzaq et al. 2013).

However, coke production is a highly energy-intensive process (Porzio et al. 2014) and is also environmentally polluting as several pollutants are emitted into the air, water and soil (Pilarczyk et al. 2013). In addition, the purification process of the by-product requires additional energy and auxiliary materials, leading to significant indirect emissions. A comprehensive life cycle assessment can help understand and identify the environmental hotspots and the potential for technological improvement in coke production and oven gas purification.

Life cycle assessment studies the environmental factors and potential impacts associated with products, from raw material sourcing through production, consumption and final disposal. It is a method that partly inventories the inputs and outputs of the system of processes related to a product, evaluates the potential environmental impacts connected to them, and interprets the results of the inventory analysis and the impact assessment phases from the perspective of the purpose of the study (Curran 2016). Due to its benefits, LCA method is widely used in different industry evaluations, for example in the construction (Essaghouri et al. 2023; Zhang et al. 2019), food (Takacs and Borrion 2020; Cai et al. 2022), chemical (Aldaco et al. 2019; Khoo et al. 2018) or iron and steel industries (Leão et al. 2023; Olmez et al. 2016; Moziraji et al. 2023).

In recent years, more and more studies have been published concerning the environmental impact and life cycle assessment of coking plants and coke oven gas utilisation.

Li et al. (2020) created a holistic cradle-to-gate LCA on coking production using the subdivision method. The results showed that COG purification contributed to air emissions of the entire coke production to a similar or even more significant than the coke production process. Concerning COG purification, the dominating contributors were ammonia recovery and blowing cooler system units due to the significant steam and electricity consumption in these units. Liu and Yuan (2016) presented a cradle-to-gate LCA to determine a Chinese-specific LCI of coke production and to compare coke wet (CWQ) and dry (CDQ) quenching methods. The results show a reduction of up to 15% in the selected environmental impact categories for coke dry quenching technology. Li et al. (2018) performed a cradle-to-gate LCA of COG-based methanol production and compared it with two other production routes. The results revealed that the coal route has the highest environmental impact among the selected categories, followed by the COG route, while the natural gas route has the lowest. It was also concluded that the COG-based route has the lowest production cost. Chen et al. (2019) performed a similar LCA study on the three types of methanol production routes. García et al. (2019) applied the LCA of the cogeneration process with waste gases (COG and Linz-Donawitz converter gas) of an integrated iron and steel plant. Five different scenarios were examined, varying the amount of waste and natural gas burned in the cogeneration process. It was found that it is important to minimise the consumption of natural gas and that waste gases are worse than natural gas in terms of CO_2 emissions, but it is still better to use them in the cogeneration process than to burn them in the torch. Li et al. (2021) conducted a comparative LCA of coke oven gas-based liquified natural gas processes and showed that the decarbonisation process has worse environmental performance than the methanation process since less product is produced. Besides, it has been proven that H_2 extraction in the methanation process is more economically desirable than without H_2 extraction. Ren et al. (2022) quantified the environmental and economic impacts of three different COG utilisation routes: COG-to-methanol, COG-to-hydrogen and COG-to-synthetic natural gas. It was presented that the COG-to-hydrogen route has the least environmental and economic impact of all three utilisation routes.

Li and Cheng (2020) created LCA to compare traditional and COG-based hydrogen production routes' carbon emissions, energy consumption and economic costs. COG-based hydrogen production has higher economic competitiveness. Li et al. (2023) conducted a comparative LCA to evaluate the environmental impacts of three different COG routes for ammonia production, including a single process and two coproduction processes (ammonia with methanol or LNG). Based on the results, the coproduction of ammonia with the methanol route has the best environmental performance of the three routes. Liu and Guo (2023) compared four COG utilisation schemes through the LCA method: COG-to-hydrogen, COG-to-natural gas, COG-to-methanol and COG-to-power. The results revealed that COG-to-power performs better in all aspects and is the most competitive utilisation scheme, while COG-to-methanol is the least competitive. Di et al. (2024) also compared different COG utilisation routes and concluded that the COG-to-synthetic ammonia route is

the preferred option for balancing environmental and economic benefits.

The previously introduced LCA studies are summarised in Table 1, showing the topics and the system boundaries.

Most of the published LCA studies in the field of coke production focus mainly on the coke oven gas utilisation potential and do not deal in detail with the analysis of the gas purification process. This study conducted a "cradle to gate"

Table 1 Literature review of LCA studies concerning coke production and COG utilisation

Source	Topic	System boundaries
Li et al. (2020)	Cradle-to-gate LCA on coking production	1. Washed coal upstream processes (coal mining, raw coal transportation, coal processing, washed coal transportation) 2. Coke- making process (coal charging, coal carbonisation, coke quenching, coke screening) 3. COG purification process (blowing cooler system, sulphur recovery, ammonia recovery, raw benzene recovery)
Liu and Yuan (2016)	Comparison of CDQ and CWQ technology	1. Raw material acquisition 2. Transportation 3. On-site coal preparation 4. Thermal distillation 5. Coke handling and storage 6. By-product recovery
Li et al. (2018) Chen et al. (2019)	COG-based methanol production, compared with coal and natural gas routes	1. Coal mining and processing, transportation, coke production, methanol production 2. Coal mining and processing, methanol production 3. Natural gas extraction, methanol production
García et al. (2019)	LCA of cogeneration process with COG and converter gas	Cogeneration process 1. Electric plant 2. Thermal plant
Li et al. (2021)	COG-based LNG process	1. Coal mining and processing 2. Coal transportation 3. Coking production 4. LNG production (decarbonisation or methanation (with/without H_2 extraction)) 5. LNG transportation 6. LNG evaporation
Ren et al. (2022)	Comparison of three COG utilisation routes	1. Coke oven gas production 2. Utilisation scenarios S- 1 Methanol production S- 2 Hydrogen production S- 3 Natural gas production
Li and Cheng (2020)	Comparison of traditional H_2 and COG-based H_2 production	1. Coal mining and processing 2. Coal transport 3. Coal to coke oven gas 4. Coke oven gas to hydrogen
Li et al. (2023)	Comparative LCA of ammonia production by COG	1. Coal mining and processing, 2. Coal transportation, 3. Coking production 4. Ammonia production (with single or methanol/LNG coproduction processes)
Liu and Guo (2023)	Comparison of four COG utilisation routes	1. Coal mining and washing 2. Coal transportation 3. Coal coking 4. Product production (hydrogen/natural gas/methanol/electricity) 5. Product utilisation to provide energy
Di et al. (2024)	Comparative LCA of four COG utilisation routes	1. Coal mining and washing 2. Coal transportation 3. Coal coking 4. COG utilisation

LCA of a hungarian coking plant to identify the crucial processes and to provide valuable insights into the environmental significance of the coke oven gas purification technology.

Furthermore, only a few research papers have been published on LCA using process simulator results for analysis concerning coke or coke oven gas production. Fu et al. (2023) used Aspen Plus software for the methanol synthesis process to provide energy, mass balance and direct CO_2 emission data for LCA. Man et al. (2014) used emissions of the coke-oven gas-assisted coal to olefins process obtained from the Aspen Plus process simulator for their LCA. There are also publications in other research areas that model complete or sub-technologies in process simulator software and use the model results for LCA (Han et al. 2019; Ye et al. 2023; Bansod et al. 2024). However, to our knowledge, few LCA study (e.g. Dias et al. 2023) has compared different working points obtained in process simulator software using LCA. Thus, besides the operational data, the optimised conditions determined from the steady-state model of the technology was also investigated using LCA and compared with the results calculated with the reference data. Therefore, the novelty of this research work is that process engineering simulation optimization has been complemented with LCA. Basically, a method was presented that can be used for operational decision making in real technologies, taking into account different aspects.

2 Process description

The investigated process is shown schematically in Fig. 1.

Coking coals are transported into the plant by bulk carriers or by rail, then waggons are emptied with the use of waggon tipplers, and coals are stored in silos before use. The different kinds of coals are blended, crushed, homogenised and charged into the chambers of the coke oven batteries. The coal blend spends 16–24 h in the airtight hot chambers. After the degasification, the coke is pushed out of the chambers and cooled by inert gas in the dry quenching technology or by water in the wet quenching technology. The absorbed heat by the inert gas in dry quenching technology is used to produce high-temperature, superheated steam in heat recovery heat exchangers, which is converted into electricity and low-pressure steam by a steam turbine. The produced electricity covers 30–40% of the plant's electricity demand, while the produced steam provides 70–80% of the steam needs. The cooled coke is graded with different coke screens and transferred to the blast furnace.

During the degasification, 45,000–60,000 m^3/h of raw coke oven gas is produced, which is cooled down by water spray directly and sucked out of the chamber via separators, electric tar separators and pre-coolers by gas-exhausters and transferred for further gas cleaning. In this process, coal tar and water are separated from the gas. Tar is a valuable by-product, while the water is used in the gas purification scrubbers, and surplus water is treated in the wastewater treatment plant by chemical and biological processes. The raw gas is transferred into the gas purification plant through a H_2S and two NH_3 scrubbers. The washing liquid leaving the scrubbers is regenerated, while the vapour fraction of the regenaration is fed to the closed Claus plant, where the heat of the crack reactor produces steam, which covers 2–5% of the steam demand.

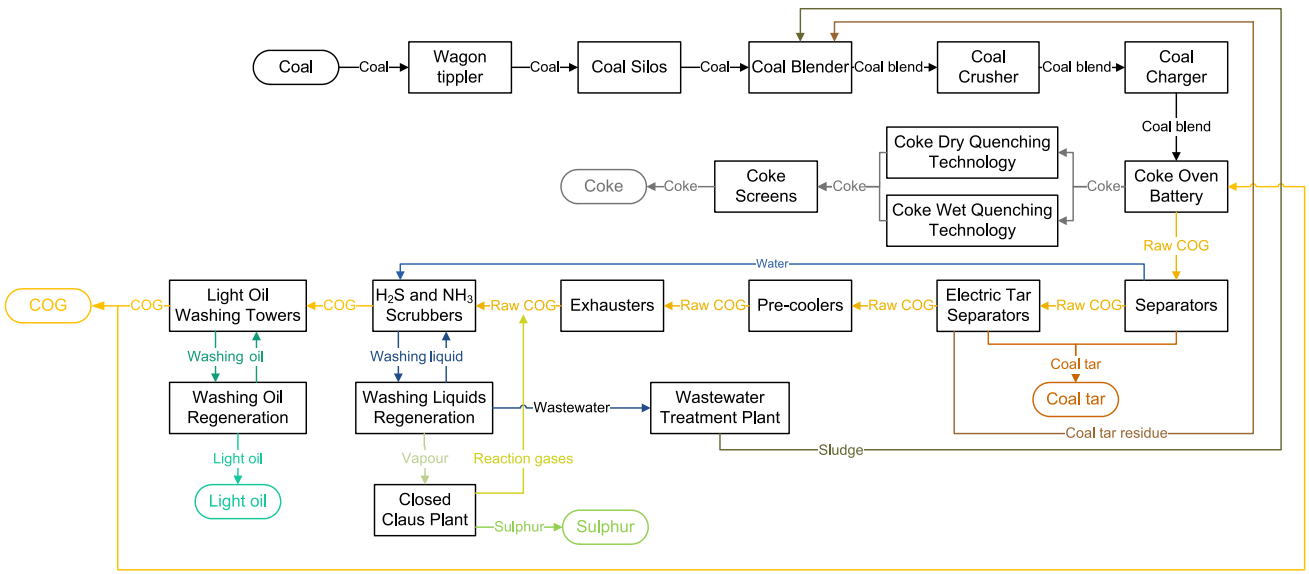


Fig. 1 Process flow diagram

After the gas purification plant, the light oil fraction is recovered from the gas in washing towers with washing oil in the light oil washing towers, and the washing oil is regenerated in distillation towers. The cleaned coke oven gas from the light oil washing towers is partly used for heating the batteries, and the remainder is used for energetic purposes.

2.1 Gas purification process and model

Gas purification is the most important process for further use of coke oven gas. For this pupose, various technologies are used, mainly absorption systems. The most commonly applied purification technology is aqueous ammonia absorption, but several technologies using more different absorbents have also emerged in this industry, such as Stretford, Takahx or Sulfiban processes (Kohl and Nielsen 1997). Further use of inadequately cleaned gas can lead to serious problems, such as exceeding environmental emission limits or increasing equipment corrosion. Therefore, this study focuses primarily on this process, and based on our previous work, different scenarios are examined.

The gas purification process can be separated into two sections: gas cleaning and regeneration. The gas cleaning section contains an H_2S and two NH_3 scrubbers, while the regeneration section has two stripping and two deacidification columns. In the scrubbers, three kinds of washing liquids are used: stripped water (the bottom product of the stripping column), de-acidified water (the bottom product of deacidification column and has high NH_3 content) and coal water (formed from the moisture content of the coal blend and separated from COG with the help of separators). When the scrubbers are not in operation due to maintenance or malfunction, the stripping columns must still be applied since the pre-treatment of the coal water is done here, and only then can it be fed to the wastewater treatment plant.

Radó-Fóty et al. (2022) developed the steady-state process model of the gas cleaning section and performed two optimisation tasks: environmental and energetic optimisation. The steady-state process model was built in Aspen Plus, and it was connected to MATLAB to accomplish the optimisation tasks. The detailed description of the model and the objective functions used are given in the referenced publication. Environmental optimisation means minimising the content of H_2S and NH_3 in the purified COG. This goal can be reached by decreasing the water stream temperature and increasing the washing liquids' flow rate. Energetic optimisation aims to reduce energy consumption and, at the same time, ensure compliance with emission standards (European Commission 2012). For this purpose, the relatively high temperature of the water streams should be used, which is why less auxiliary energy is required.

The results of the two optimization tasks were used in this study for scenario 2 and scenario 3. The calculated results, namely the amount of washing liquids and the composition of the purified coke oven gas, were used as input parameters for the life cycle assessment in Sphera's Life cycle assessment software.

3 Materials and methods

This study was conducted by ISO 14040 (International Standard Organization 2006a) and ISO 14044 (International Standard Organization 2006b) standards, so it includes four interrelated phases:

- Definition of goal and scope
- Life cycle inventory analysis (LCI)
- Life cycle impact assessment (LCIA)
- Life cycle interpretation (this phase is discussed in the 4. Results and discussion section)

3.1 Goal and scope

Defining the goal and scope is the first phase at the beginning of the study, before data collection, and it is an essential part of LCA methodology since it is where the accurate approach to be followed is determined (Curran 2017).

This study aimed to assess the environmental impact of coke production and compare four different scenarios. The four scenarios are as follows: normal operating conditions (scenario 1), environmentally optimised conditions of the gas purification process (scenario 2), energetically optimised conditions of the gas purification process (scenario 3) and not operating gas purification process conditions (scenario 4).

Based on Fig. 2, the boundaries of the system were defined. Thus, the previously introduced processes were considered in eight sub-processes: transportation, loading and storage, coking process, coke quenching and screening, cooling and tar recovery, gas purification, wastewater treatment and light oil recovery. Malfunctions and accidents were not taken into account in the assessment.

For each sub-processes, the energy and auxiliary flows were modelled using country-specific (e.g. electricity grid mix, process steam from natural gas) or, if not available in the database, European data (e.g. compressed air).

Transportation: Only coal transportation was considered in this sub-process due to the long distance. While the used auxiliary and other raw materials are purchased relatively close to the plant, and the products are used locally or sold nearby, transporting these materials is negligible compared

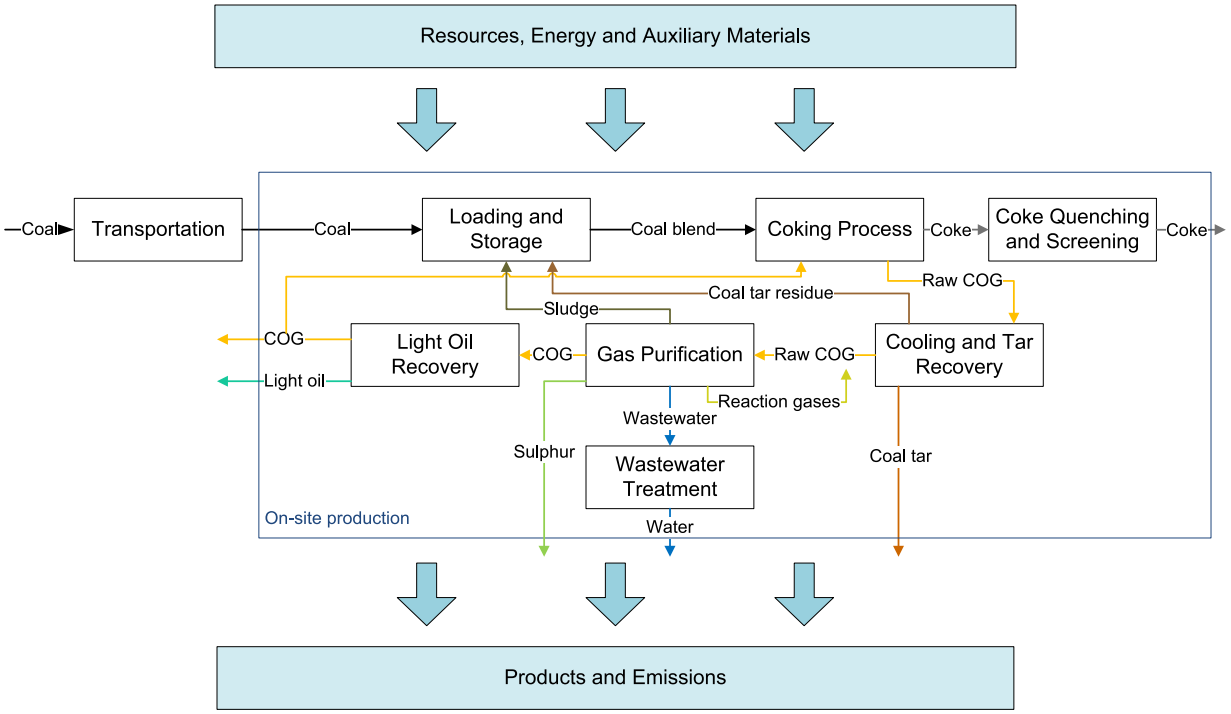


Fig. 2 System boundary of the process

to coal. Forty percent of the used coals are transported from other continents with an average distance of 11,500 km by bulk carriers, modelled by "GLO: Bulk commodity carrier. average. ocean-going". The rest of the coal used is delivered to the plant from European countries by rail, from an average distance of 726 km. This form of transportation was modelled by "GLO: Rail transport cargo—Electric. average train. gross tonne weight 1.000 t/726 t payload capacity". Electricity and heavy fuel oil are consumed during these two kinds of transportation.

Loading and storage: The coal-filled waggons arriving at the plant are tippled, loaded into silos, blended, crushed and homogenised. These processes operate in a closed system and use dust extraction equipment for those stages that emit dust (e.g. crushing) so that dust emissions into the air are minimised. The dust removed in the separators is added to the coal blend. In addition, sludge from the wastewater treatment plant is mixed with the coal blend. The residues from the tar separation (coal tar residue and coal tar pitch residue) are also blended into the coal blend.

Coking process: This sub-process consists of two coke oven batteries and the coking machines responsible for coal loading (charging car) and coke pushing (pusher machine and coke guide car). Coke pushing emission control system limits the release of dust during chambers are pushing. The cleaned coke oven gas is used in excess air to heat the coke oven batteries, which causes emissions (e.g. NO_x, SO_2, CO_2, particulate).

Coke quenching and screening: Two types of coke quenching, dry and wet quenching and coke screening, were considered in this stage. Dry quenching, as described previously, produces electricity and steam, reducing the technology's energy purchase demand. The coke dust generated during coke classification is separated and mixed back into the coal blend. As the quantity of coke dust is minor compared to the coal used, it is not specially accounted for in the analysis.

Cooling and tar recovery: The raw coke oven gas is cooled and separated in a closed system, so under normal operating conditions, no pollutants can be released into the air, water or soil. The residues of this process are mixed with the coal blend, but this is negligible compared to the total amount of coal used and is therefore not taken into account in this study.

Gas purification: This sub-process is the most important for this study. Like cooling and tar recovery, it is a closed system, too, so no direct emissions can occur. In this stage, the largest amount of auxiliary material, sodium hydroxide (2.5 kg/ton of coke), is used to regenerate the washing liquids. Due to the closed Claus technology, in

the cracking reactor, the parameters are adjusted so that the residual reaction gases at the end of the process can be recycled into the coke oven gas system. This means residual gases do not need to be burned in a flare, and no flue gas is emitted.

Wastewater treatment: The surplus process water is treated at the wastewater treatment plant before being discharged to the receiving water. The discharged water is regularly monitored to ensure that the emissions (e.g. nitrate, BTEX, BOD_5, PAH, sulphide, cyanide) are below environmental standards. Various auxiliary materials are used for wastewater treatment (e.g. flocculant, defoamer, hydrated lime, glucose), the amount of which is below 1 kg/ton of coke, but to describe the process more precisely, it has been taken into account.

Light oil recovery: This is the final stage of the gas treatment process. The quality of the washing oil used for light oil recovery is modified by the polymerisation reactions that occur during heating. These changes are technologically highly negative because, on the one hand, the substances precipitated by polymerisation can cause blockages in the pipes, and, on the other hand, the ability of the oil to absorb benzene is reduced. In order to avoid this, the washing oil must be regenerated. The residual polymerised oil is fed into the tar recovery process.

Coke production process is a multi-output (or multi-functional) system with five valuable products; thus, allocations of environmental burdens are required. According to Li et al. (2023), mass allocation method was used, as it is one of the most commonly applied methods. Furthermore, mass allocation can be reliably applied to all products, unlike energy allocation, as sulphur is not an energy substance, while economic allocation cannot be used due to lack of information. The mass allocation factors ($C_{\mathrm{mass,i}}$) are collected in Table 2 and based on Eq. (1), where m_i is the mass of a product, and m_{total} is the mass of all the five products.

$$C_{mass,i} = \frac{m_i}{m_{total}} \tag{1}$$

3.2 Life cycle inventory (LCI)

Life cycle inventory is a process to quantify the inputs (e.g. energy, resources and raw materials) and outputs (e.g. products, emissions and residues) for a product system and includes the necessary data collection and calculation procedures (Finnvenden and Potting 2014). High-quality data is necessary to guarantee the reliability and completeness of the LCA study.

All the primary data used, such as the amount of raw and auxiliary materials, energy, water and transportation distance, were obtained from a hungarian coke oven plant with a capacity of 1.2 million tons of coke. The quantities of air emissions (CO_2, NO_x, SO_2 and particulate matter) were collected from annual environmental declarations. The overall inputs and outputs data of coke production referring to 1 ton of coke in the case of scenario 1 are collected in Table 3.

Table 2 The allocation factors of each product

Product	Mass (kg)	$C_{\mathrm{mass,\,i}}$ (%)
Coke	1000.00	78.78
Coal tar	43.30	3.41
Light oil	12.60	0.99
COG (purified)[a]	212.04	16.71
Sulphur	1.35	0.11

[a]The density of COG is 0.46 kg/Nm^3

In order to show the distribution of energy and water used in the sub-processes (transportation was not considered relevant in this case), detailed data for scenario 1 are listed in Table 4. It can be observed that the largest electricity consumer is cooling and tar recovery, followed by gas purification and then quenching and screening. Gas purification uses a major amount of steam and also has a significant process water demand. Desalinated water is used for steam production. Nitrogen is mainly applied in places with fire and explosion hazards and is the inert gas in coke dry quenching technology.

Background data, for example, shows indirect emissions from electricity, steam, water and different chemicals; furthermore, the transportation models were insured by Sphera's life cycle assessment software (formerly and commonly known as GaBi) database (Sphera 2023).

For the other three scenarios, the main differences in inputs, outputs and emissions are summarised in Table 5, which relates to gas purification and coking process. Data not included in Table 5 are the same as in Table 3 for all scenarios. In case of scenario 4, the unpurified coke oven gas is partly used for heating coking chambers as there is no other way to heat them, but the surplus cannot be traded because of the high concentration of impurities, so this amount is burnt in a torch.

3.3 Life cycle impact assessment (LCIA)

Despite the fact that much knowledge can be gained about a process by examining the LCI data, an impact assessment represents a more meaningful basis for making comparisons. The LCIA uses science-based characterisation factors, which help calculate each environmental emission's

Table 3 The overall inputs and outputs of coke production refer to 1 ton of coke (scenario 1)

Parameter	Amount	Unit
Input		
Coal blend	1.39	t
Washing oil	0.94	kg
COG for heating	4.30	GJ
Electricity (purchased)	0.24	GJ
Electricity (produced)	0.11	GJ
Steam (purchased)	0.33	GJ
Steam (produced)	1.73	GJ
Process water[a]	3.57	m^3
Drinking water	0.88	m^3
Desalinated water	0.13	m^3
Nitrogen gaseous	3.62	m^3
Compressed air	33.94	m^3
Sodium hydroxide	2.50	kg
Acetic acid	1.47	g
Defoamer	18.66	g
Flocculant	56.48	g
Glucose	388.51	g
Hydrated lime	84.73	g
Phosphoric acid	1.72	g
Sodium hypochlorite	81.04	g
Sulphur	42.36	g
Output		
Coke	1.00	t
Coal tar	43.30	kg
Light oil	12.60	kg
COG for trading	3.74	GJ
Sulphur	1.35	kg
Emissions to water		
Wastewater	0.59	m^3
Nitrate	6.90	g
Total nitrogen	16.00	g
BTEX (benzene, toluene, ethylbenzene, xylenes)	4.62	mg
PAHs (polycyclic aromatic hydrocarbons)	0.44	mg
Sulphide	0.06	g
Cyanide	0.01	g
Phosphorus	0.11	g
BOD_5 (biochemical oxygen demand, 5-day)	1.77	g
Emissions to air		
Particulates	0.12	kg
CO_2	459.5	kg
NO_x	1.19	kg
SO_x	1.21	kg

[a]Process water contains mainly cooling water

potential impact on problems such as global warming. The LCIA phase evaluates the potential impacts on human health and the environment of the environmental resources and emissions identified during the inventory (Curran 2016).

The four LCA models were implemented in Sphera, and the LCA evaluation was made using the CML 2001 method. This method was developed by the Center of Environmental Science of Leiden University (Centrum voor Milieukunde Leiden: CML) in the Netherlands in 2001. For the assessment of environmental impact, a wide range of characterisation methods and impact categories were established (Dincer et al. 2018).

In this study, nine related midpoint environmental impact categories, abiotic depletion (ADP fossil), acidification potential (AP), eutrophication potential (EP), freshwater aquatic ecotoxicity potential (FAETP inf.), global warming potential (GWP 100 years), human toxicity potential (HTP inf.), marine aquatic ecotoxicity potential (MAETP inf.), photochemical ozone creation potential (POCP) and terrestric ecotoxicity potential (TETP inf.) were considered.

4 Results and discussion

4.1 LCA results of coke production based on scenario 1

The LCA results of the coke production process are presented through the results of scenario 1, as it reflects the normal operating conditions of a hungarian coke oven plant. Figure 3 shows the relative contribution of each sub-process to the nine previously introduced environmental impact categories to explore the individual contribution of each sub-process.

For five of the nine selected categories, the coking process exhibits a significantly higher impact than the other sub-processes. The coking process has the most significant impact on GWP (85%), followed by AP (80%), POCP (74%), EP (63%) and finally HTP inf. (28%). In the other categories, the impact of the coking process is below 10%. The reason is mainly attributed to the pollutant emission caused by the combustion of coke oven gas (Wang et al. 2018; Cheng et al. 2024).

Transportation has a remarkable contribution to all nine categories, but the largest percentage can be observed for FAETP (41%). This can be explained by the fact that almost half of the coking coal used comes from other continents via bulk carriers. Marine transportation is still a seriously environmentally polluting form of transportation, causing air pollution (greenhouse gases, volatile organic compounds, particle matter, ozone-depleting substances and sulphur oxides) and water pollution (release of ballast water that contains aquatic invasive species, use of antifoulants, spills and operational discharges of cargo and oil, wastewater and marine litter emission underwater noise) (Jägerbrand et al. 2019; Walker et al. 2019).

The contribution of cooling and tar recovery and gas purification is approximately similar in all categories. The

Table 4 Distribution of input auxiliary streams (scenario 1)

	Electricity (%)	Steam (%)	Compressed air (%)	Nitrogen (gaseous) (%)	Process water (%)	Drinking water (%)	Desalinated water (%)
Loading and storage	3	3	28	0	0	16	0
Coking process	7	5	23	0	1	24	0
Quenching and screening	23	18	30	17	4	19	98
Cooling and tar recovery	37	16	7	36	19	14	0
Gas purification	24	30	7	23	32	13	2
Wastewater treatment	5	1	3	0	43	2	0
Light oil recovery	1	27	2	24	1	12	0

Table 5 The main differences between the four scenarios refer to 1 ton of coke

Parameters	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Unit
Gas purification					
Input					
Electricity (purchased)	60.48	81.94	39.74	12.20	MJ
Process water	1.16	1.36	1.01	0.01	m^3
Output					
COG for trading	3.74	3.74	3.74	0	GJ
Sulphur	1.35	1.38	1.31	0	kg
Water to wastewater treatment	0.60	0.91	0.37	0.14	m^3
Coking process					
Main emissions					
NO_x	1.19	1.18	1.39	2.27	kg
SO_x	1.21	1.08	1.29	3.55	kg

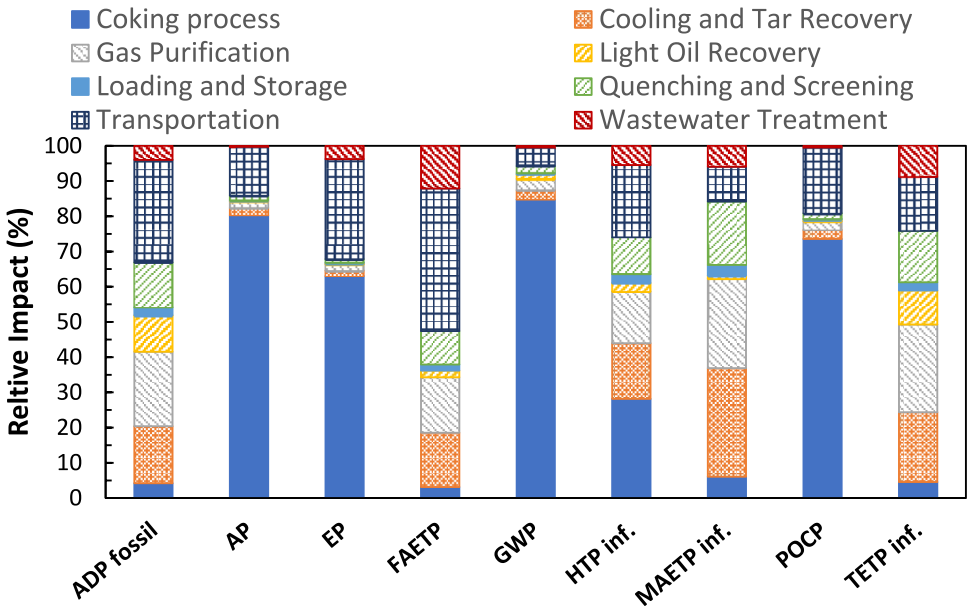


Fig. 3 LCA results of coke production based on scenario 1

contributions of ADP fossil, FAETP, HTP inf., MAETP inf. and TETP inf. are more prominent with 15–30% of the total, compared to only 2–3% for the other categories. A similar tendency can be observed in the case of quenching and screening, but the contribution rate of the previously mentioned five categories is only 9–18%.

Despite the fact that coking wastewater is known to contain a variety of organic and inorganic pollutants (Wang et al. 2024), the results show that the environmental impact of this sub-process can be minimised by appropriate treatment technology. It is seen that wastewater treatment contributes 12% and 9% in FAETP and TETP inf., respectively, and below 6% for the other categories. The light oil recovery process contributes only to ADP fossil and TETP inf. (10%), while the contribution of the other categories is almost negligible (0.1–2%). The process of loading and storage has the lowest environmental impact, with a contribution below 3% for all categories, because the energy consumption is comparatively low, the emissions are not significant due to the closed structure of the conveyor belts and silos, and the dust collectors above coal crushers.

Furthermore, Fig. 4. depicts the relative impact of each input auxiliary stream on each environmental impact category. The relative contribution of drinking water, nitrogen, desalinated water and compressed air is less than 1% for each environmental impact category. Electricity has the largest relative contribution for AP (85%), EP (57%), FAETP (61%), HTP inf. (74%), MAETP inf. (93%) and POCP (73%). Steam contributes most significantly to ADP fossil (58%), GWP (48%) and TETP inf. (46%). The impact of process water is less significant (all categories are below 25%) than electricity or steam but still not negligible. It has the highest contribution to EP (21%) and FAETP (25%).

4.2 Results of the comparative LCA

The results of the four scenarios are compared in this section. Since, for the different scenarios, significant changes can only be observed in the coking process and the gas purification, the results of each scenario are presented focusing specifically on these sub-processes, while the complete LCA results for each scenario are shown in Figs. 11, 12, and 13 in the Appendix.

Figure 5 presents the relative contribution of the coking process to the nine environmental impact categories in the four scenarios. It can be noted that for all categories, compared to scenario 1, scenario 2 decreases, while scenarios 3 and 4 increase. Since, for scenario 2, the conditions under which the amount of impurities in the coke oven gas is minimal, less polluting components are emitted into the environment during combustion. In the energetically optimised case (scenario 3), the concentration of pollutants in the gas is still below the environmental limit (in coke oven gas, the concentration of $H_2S < 1$ g/Nm^3 and the concentration of $NH_3 < 0.1$ g/Nm^3 (European Commission 2012)), but the environmental impact of the process is slightly increased compared to the base case (scenario 1). When there is no gas purification at all (scenario 4), which means that the concentration of pollutants in the combusted coke oven gas is significantly above the standards, a spectacular increase in the relative impact can be observed, especially for AP, EP, HTP inf. and POCP. Figure 6 illustrates the relative difference between scenario 1 and the other scenarios. Scenario 2 shows the largest decrease for MAETP inf (− 7%). While for both scenario 3 and scenario 4, the largest increases are observed for HTP inf. (13% and 64%) and MAETP inf. (7% and 27%).

The environmental impact of the other relevant subprocess, gas purification, is shown in Fig. 7. Compared to scenario 1, all categories show an increase for scenario 2, while scenario 3 and scenario 4 show a decrease. In scenario 2, the electricity demand of the process increased (the

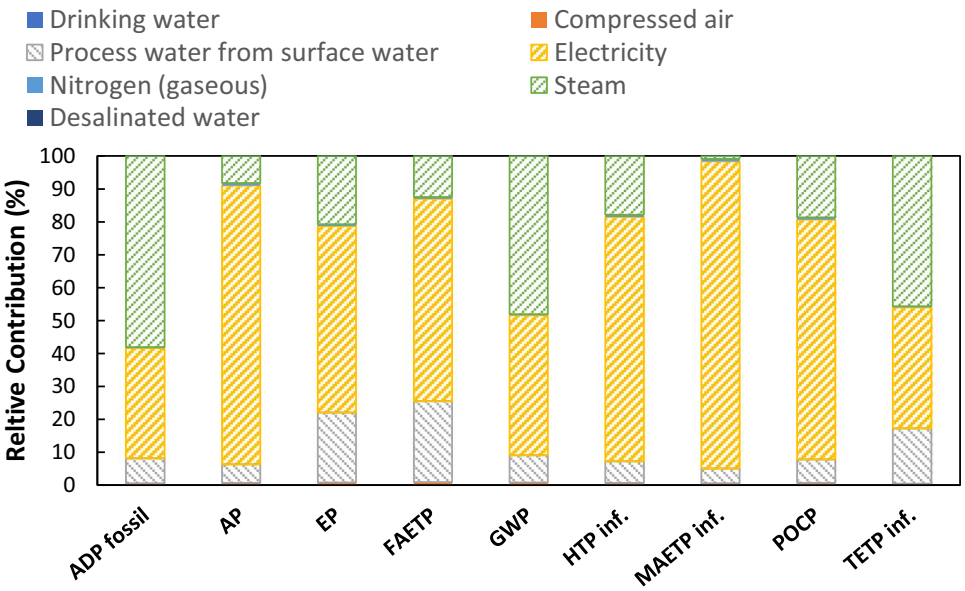


Fig. 4 Relative impact of input auxiliary streams based on scenario 1

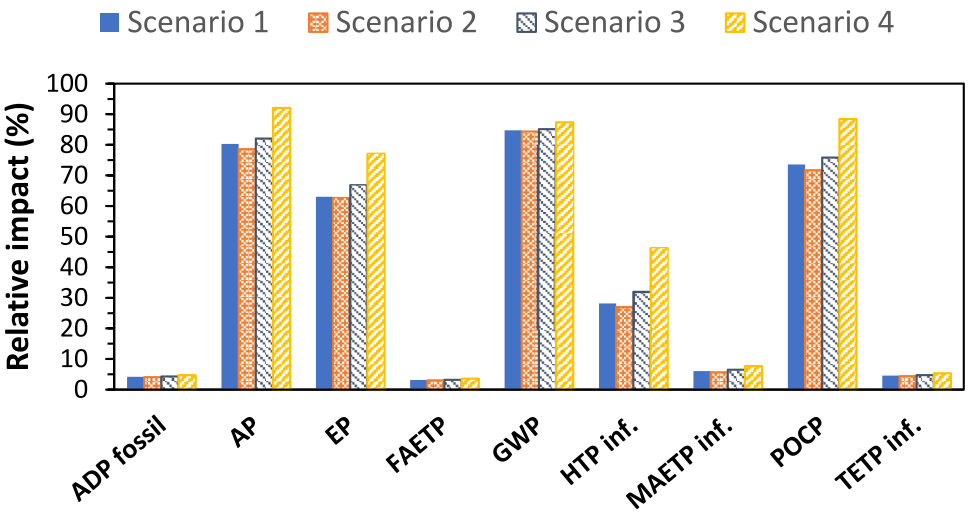


Fig. 5 Relative impact of coking process for the four scenarios

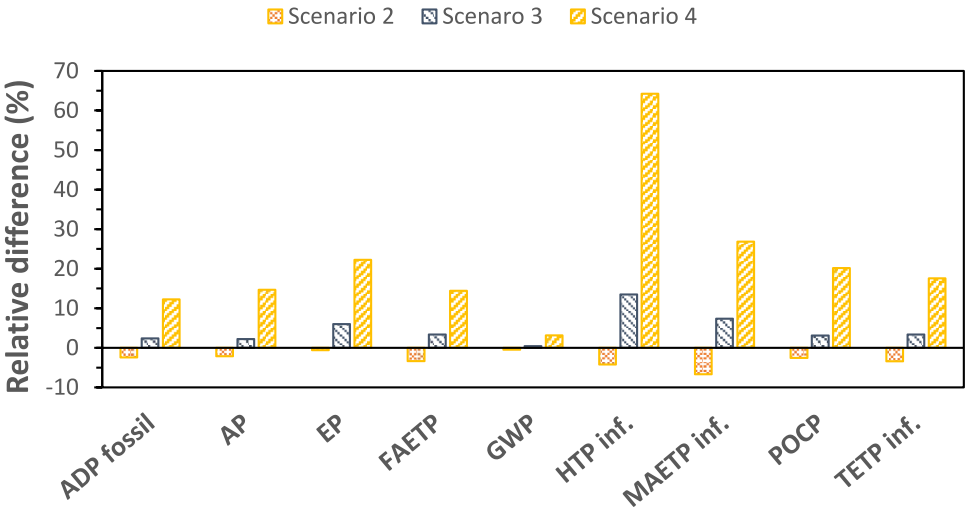


Fig. 6 Relative differences in environmental impact categories compared to scenario 1 of the coking process

washing liquids have to be cooled to lower temperatures), which increased the relative impact in each environmental category. In scenario 3, on the other hand, the energy demand of the process was minimised, so the environmental impact was also reduced. In scenario 4, the scrubbers are not in operation, only the stripping columns are in use, which is a small part of the sub-process, meaning that significantly less energy and auxiliary materials are consumed.

Figure 8 illustrates the relative difference between scenario 1 and the other 3 scenarios. For scenario 2, the most significant increase can be observed for AP (33%) and POCP (28%). In scenario 3, on the other hand, these two categories showed the largest decreases, with AP decreasing by 30% and POCP by 27%. Scenario 4 also showed a similar trend but a much larger decrease, with AP decreasing by 91% and POCP by 88%.

Figure 9 shows the relative differences between scenario 1 and the other scenarios, considering the whole technology. It can be seen that the most drastic difference is between scenario 1 and scenario 4. Among the environmental categories examined, increases of 130%, 113%, 56%, 13% and 7% were observed for AP, POCP, EP, HTP inf. and GWP, respectively. In contrast, the decreases for ADP fossil, FAETP, MEATP inf. and TETP inf. range from 10–21%. In scenario 2, only three categories (AP, EP and POCP) show small decreases (0.1–6%), but the increases in the other categories are also modest, ranging from 0.5 to 7%. For scenario 3, there were reductions in most categories, including ADP fossil, FAETP, GWP, MAETP inf. and TETP inf., although the reduction rate ranged from only 0.4–7%. The other four categories showed increases but at a much more modest rate than for scenario 4, ranging from 0.8 to 10%.

Table 6 shows the absolute values of the four scenarios in AP and GWP impact (while Table 7 contains the absolute values for every impact category). In the case of AP, it can be observed that the lowest value belongs to the environmentally optimised conditions (scenario 2), as the concentration of

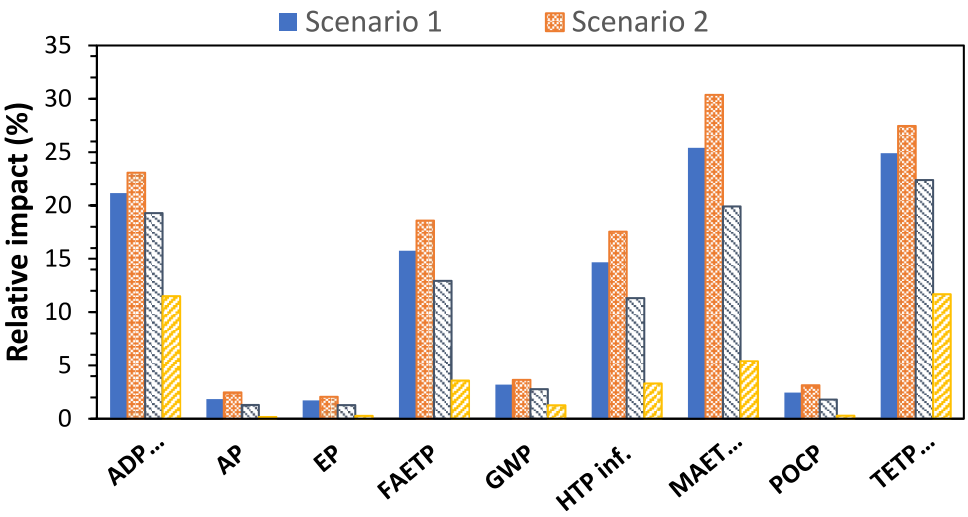


Fig. 7 Relative impact of gas purification for the four scenarios

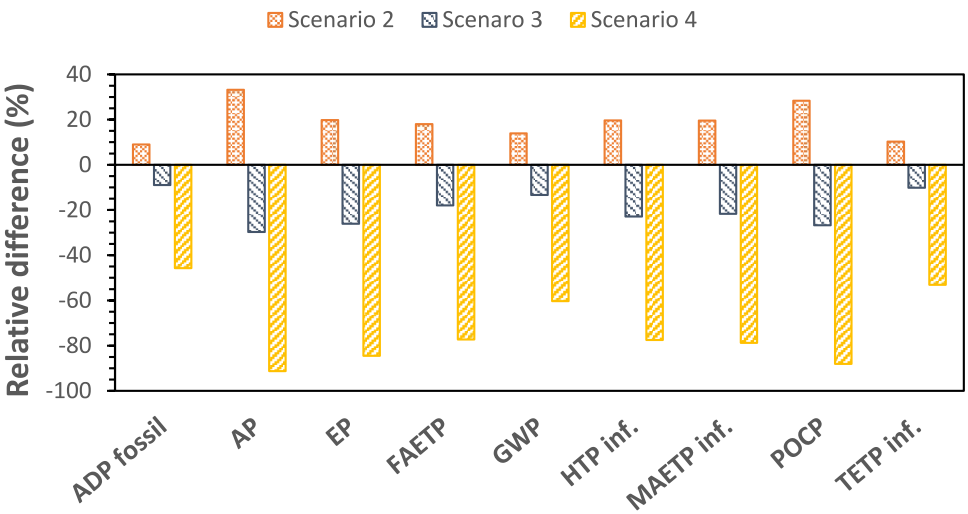


Fig. 8 Relative differences in environmental impact categories compared to scenario 1 of the gas purification

H_2S in coke oven gas is the lowest that can be achieved with the existing gas purification technology. In contrast, if there is no gas purification (scenario 4), this value increases by almost 2.5 times due to the impurities in the gas. The lowest value for GWP, however, belongs to the energetically optimised conditions (scenario 3), as the energy demand for gas purification is the lowest. While the highest value in this case is also related to scenario 4, due to the remaining polluting components in the coke oven gas.

Furthermore, Table 6 is supplemented with results of two LCA studies to verify the robustness and validity of the LCIA results. The results of this study are consistent, and there is a relatively good agreement between the results of scenarios 1–3 and PE-GaBi in the case of AP, while the results of GWP are close to the results of Li et al. (2020) with subdivision. This difference is partly due to the different methods, as it seems that the GWP is closer to the subdivision method of Li et al. (2020) than to the method without subdivision, as this study also used subdivision method. However, the sensitivity analysis also shows that changes in technological parameters have a significant effect on the environmental impact categories. This means that there are both technological differences (e.g. Liu and Yuan (2016) compared CDQ and WDQ technologies separately, while in this study, both technologies are used in the same time) and differences in input and output parameters (e.g. the amount of coal blend needed to produce 1 ton of coke is also different, which may be due to the different composition of the coal blend) in the referenced literatures in the comparison.

4.3 Results of sensitivity analysis

The rising cost of energy is increasing the focus on energy efficiency of plants. For this reason, a sensitivity analysis was

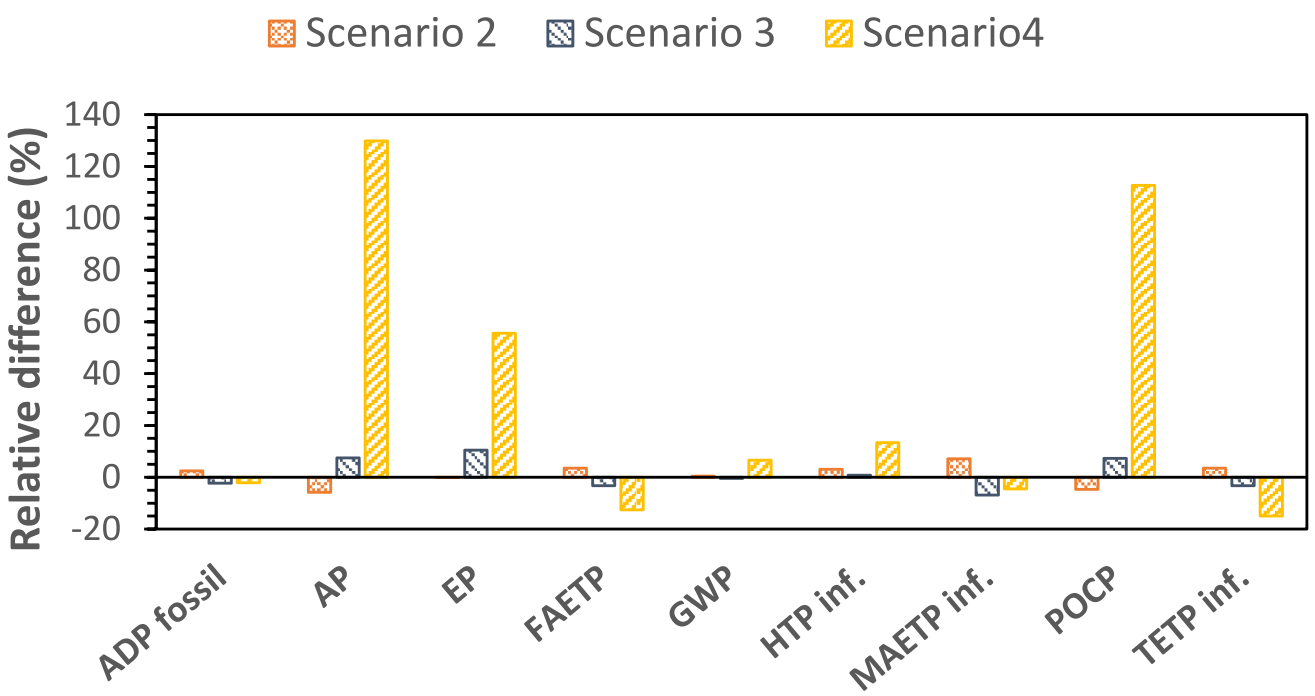


Fig. 9 Relative differences in environmental impact categories compared to scenario 1, considering the whole technology

Table 6 Comparative analysis of coke production in AP and GWP

		AP (kg SO_2 eq.)	GWP (kg CO_2 eq.)
Scenario 1		2.56	546.54
Scenario 2		2.41	549.07
Scenario 3		2.75	544.15
Scenario 4		5.87	582.42
Li et al. (2020)	Without subdivision	1.22	615.10
	With subdivision	0.63	520.90
Liu and Yuan (2016)	Scenario 1 (CWQ)	8.51	1670.00
	Scenario 2 (CDQ)	7.30	1570.00
	Ecoinvent	4.74	660.00
	PE-GaBi	2.43	750.00

carried out to investigate how the environmental impacts of the current base case would change if the electricity demand of the sub-processes were reduced by 10%. Transportation was not included in the sensitivity analysis, since its environmental impacts are constant as the energy improvements only affect the operational parts of the plant.

Figure 10 shows the results of the sensitivity analysis in terms of the relative differences compared to the base case. It can be concluded that there is a significant decrease in all impact categories. In terms of impact categories, the largest reductions were for MAETP inf., AP, POCP and FAETP. These are due to electricity production. On average, a relative difference of over 5% can be observed for the following sub-processes: cooling and tar recovery, quenching and screening, loading and storage and gas purification. The most energy-intensive parts of the plant have the largest reduction in environmental impact. Thus, the results also show that significant reductions can be achieved through energy optimisation, investments and upgrades (e.g. pumps or engines with better efficiency).

5 Policy implications

Although coke production releases significant pollutants and is a rather energy- and CO_2-intensive industry sector, it is an indispensable raw material for the iron and steel industry. In addition, several coke producers, particularly small- and medium-sized companies, tend to have low environmental awareness and show a lack of environmental responsibility. Therefore, it is important that environmental authorities and policy decision-makers establish appropriate restrictions on plants that do not make their operation impossible but set reasonable limits on the operating conditions.

In the present economic situation and environmental regulations, multi-objective optimisation should be preferred when designing new plants and developing and operating existing ones. In this paper, it is presented that the

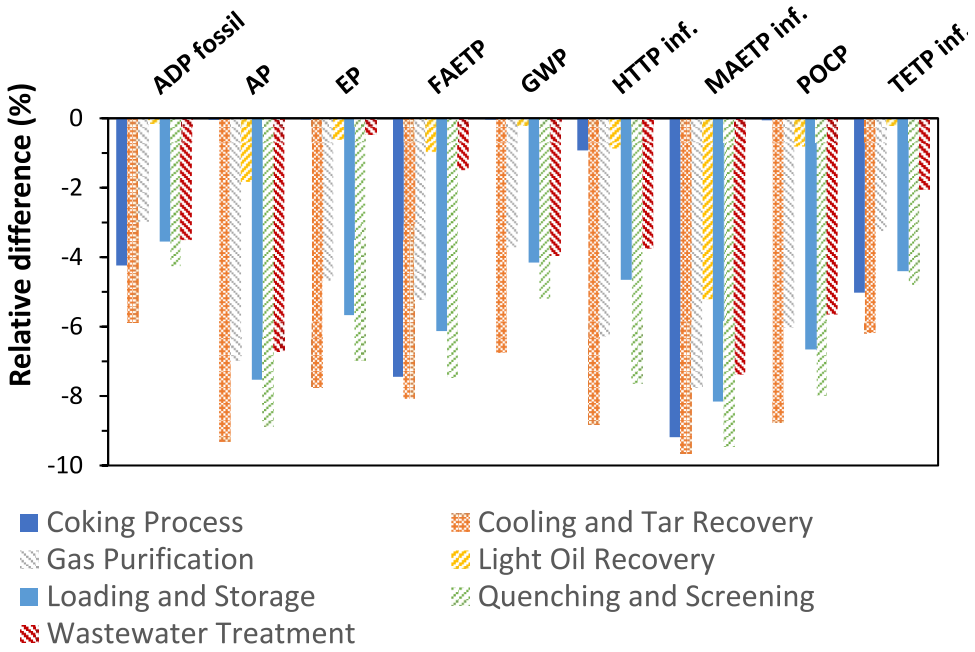


Fig. 10 Results of the sensitivity analysis of energy demand compared to the base case (scenario 1)

optimisation using process simulators can be complemented by life cycle analysis, providing a more sophisticated tool for process optimisation and thus proposing industry solutions for the energetic improvement of existing technologies.

The methods described above are transferable to similar technologies. The environmental impact of each sub-process can be examined, highlighting opportunities for improvement, which sub-processes should be given greater emphasis, and which input energy sources and auxiliary materials can be reduced to help reduce the environmental impact of the technology, keeping in mind company policies.

6 Conclusions

This study assessed the environmental impacts of a by-product recovery coke production process, focusing mainly on coke oven gas purification. LCA based on process simulation results is particularly effective in supporting the sustainable operation of existing plants in the context of dynamically changing energy prices and tightening environmental regulations. Identifying the environmental impact of critical process stages and state variables during technology development allows to integrate the response to unexpected external effects into decision processes. By analysing the technology in detail, different impacts can be modelled in advance, and environmental impacts can be determined at each stage of the process.

Based on the results of the base case scenario 1, the coking process had the largest impact in several categories (AP, EP, GWP, HTP inf. and POCP). In addition, transportation, gas purification and cooling and tar recovery sub-processes had significant environmental impacts. It can be concluded that electricity had the largest contribution to the environmental impact categories among the input auxiliary materials, followed by steam and process water.

The results of the basic assessment were compared with the results of three different scenarios (scenario 2, environmentally optimised gas purification; scenario 3, energetically optimised gas purification; scenario 4, no gas purification). In the coking process, the relative impact decreased for every category in scenario 2 (between 0.6 and 6.7%), while it increased in the other two scenarios (2.2–13.5% for scenario 3 and 3.1–64.2% for scenario 4) because the composition of the coke oven gas used to heat the chambers changed. In contrast, the opposite trend can be observed for gas purification, as scenario 2 requires more energy investment to clean the gas properly, increasing the process's environmental impact. The growth for scenario 2 varied between 9.0 and 33.2%. However, as the gas purification rate decreased or was eliminated, less energy was needed for the plant part. Thus, a reduction of between 8.9 and 29.7% was observed for scenario 3 and between 45.7 and 91.3% for scenario 4.

Comparing the environmental impacts of the whole technology for the different scenarios, it can be observed that, compared to the base case, there is a reduction in 3 categories for scenario 2, 5 categories for scenario 3 and 4 categories for scenario 4. The largest increase is obtained in scenario 4 for AP (130%), EP (56%) and POCP (113%). The comparison of the results indicated that the energetically optimised condition is the most favourable scenario in terms of environmental impacts also. The reason for this is that the environmental impact categories increase by 10% or less, while in the other categories, the decrease ranges from 2 to 7% leading to an overall better performance.

Appendix

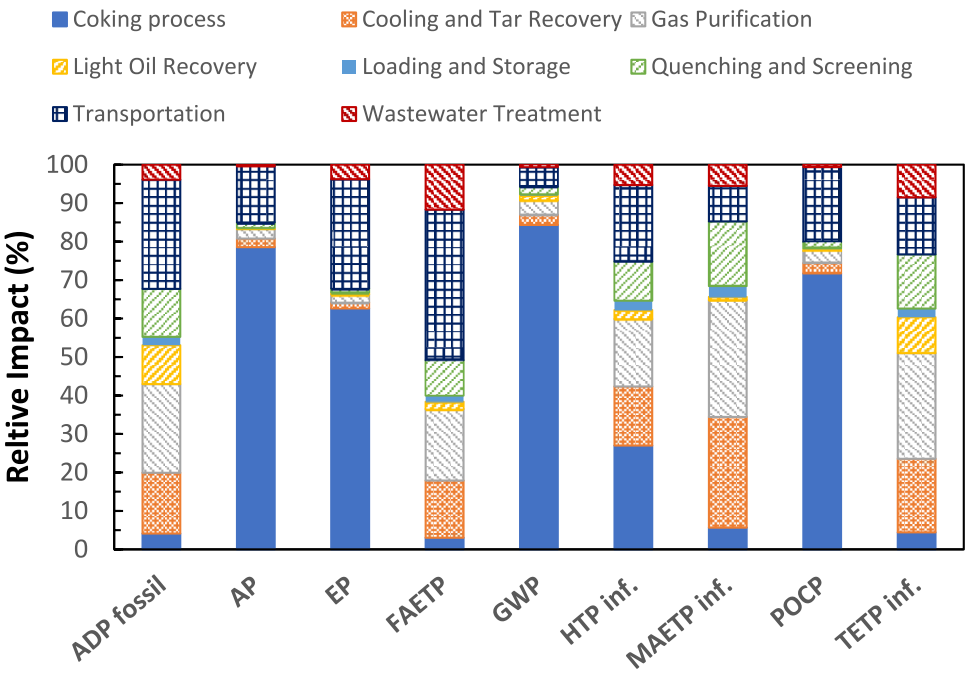


Fig. 11 LCA results of coke production based on scenario 2

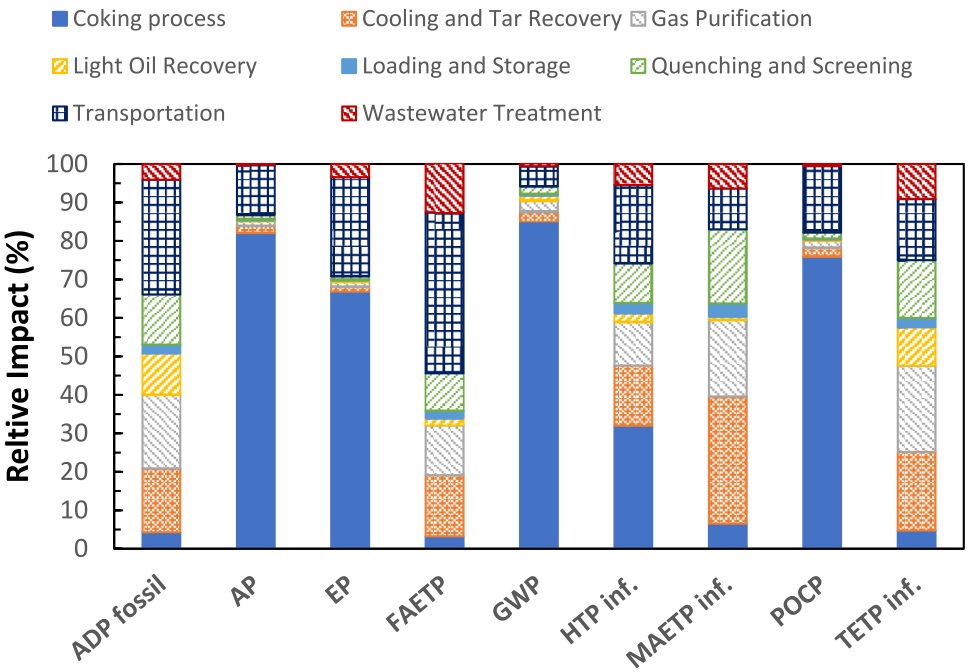


Fig. 12 LCA results of coke production based on scenario 3

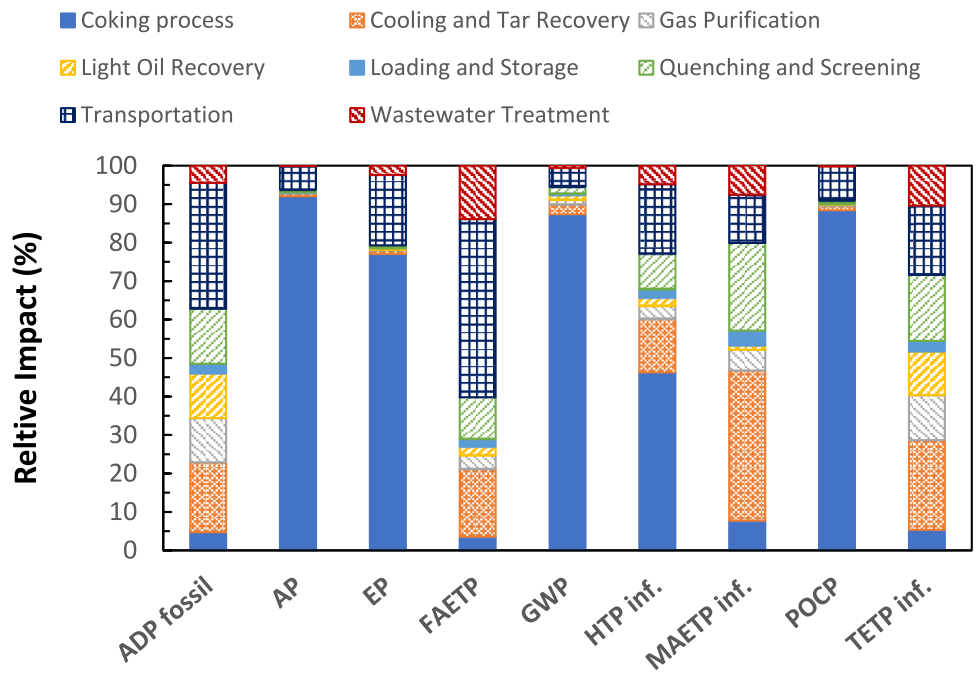


Fig. 13 LCA results of coke production based on scenario 4

Table 7 LCA results of coke production for 1 ton of coke

Impact	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Unit
ADP fossil	11,360.27	11,641.64	11,094.80	11.118,90	MJ
AP	2.56	2.41	2.75	5.87	kg SO_2 eq
EP	0.25	0.25	0.27	0.38	kg Phosphate eq
FAETP	0.25	0.26	0.24	0.22	kg DCB eq
GWP	546.54	549.07	544.15	582.42	kg CO_2 eq
HTP inf	6.26	6.45	6.31	7.09	kg DCB eq
MAETP inf	11,999.29	12,855.27	11,175.97	11,461.03	kg DCB eq
POCP	0.12	0.12	0.13	0.27	kg Ethene eq
TETP inf	0.16	0.17	0.16	0.14	kg DCB eq

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Data availability The data presented in this manuscript are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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